



BNPL (Buy Now Pay Later) Risk Scoring & Analytics (India)

A Data-Driven Approach for Customer Risk Assessment & Business Analytics

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DATA SCIENCE INTERNSHIP PROJECT

Agenda

Presentation Roadmap

This presentation demonstrates the complete workflow followed to analyze customer behavior, identify repayment risks, dashboarding & variable analysis for business decision-making.

Agenda

- Introduction to BNPL
- Business Problem & Project Objectives
- Dataset Overview & Project Methodology
- Analysis Summary (SQL, Statistics & Python EDA)
- Machine Learning Analysis
- Power BI Dashboard Analysis
- Competitor Analysis
- Business Insights & Future of Project
- Strategic Recommendations
- Conclusion.

Introduction & Business Problem

What is BNPL?

Buy Now Pay Later (BNPL) is a digital credit solution that allows customers to purchase products instantly and repay later through flexible installments. With the rapid growth of fintech and digital payments in India, understanding customer risk has become essential for sustainable lending.

Business Problem

- Rapid BNPL adoption has increased repayment defaults and financial risk.
- Traditional credit scoring alone cannot accurately predict customer behavior.
- Financial institutions require data-driven insights for better lending decisions.

Project Objectives

- Analyze customer demographics and borrowing behavior.
- Study repayment patterns and default risk.
- Compare the performance of major BNPL companies.
- Develop an interactive Power BI dashboard.
- Generate business insights and strategic recommendations.

Dataset Overview & Project Methodology

Dataset Overview

The project uses a synthetic BNPL dataset containing customer demographics, financial information, loan details, repayment behavior, and default risk indicators.

Dataset Highlights

- 70,000K+ Customer Records
- 16+ Business Features
- 4 Major & Other Minor Lending Companies
- Multi-year customer transaction data.

Key Attributes

- Age • Monthly Income • Loan Amount • Credit Limit • Credit Utilization Ratio • Repayment Method • Missed Payments • Default Risk • Lending Company.

Project Workflow

 Dataset Cleaning →  SQL Analysis →  Statistical Analysis →  Python EDA →  Machine Learning →  Power BI Dashboarding →  Business Insights.

Analysis Summary

Tools & Techniques Used

- **SQL (DuckDB)**: Customer profiling, loan analysis, repayment trends, and company-wise analysis.
- **Statistical Analysis**: Descriptive Statistics, Gaussian Distribution, Correlation Matrix, T-Test, and Z-Test to validate data patterns and relationships.
- **Python (Pandas, NumPy, Matplotlib & Seaborn)**: Data cleaning, feature engineering, and exploratory data analysis (EDA).

Key Findings

- **Gen Z** emerged as the largest BNPL customer segment, contributing the highest transaction activity and loan demand.
- **Slice & Simpl** dominated the lending portfolio, while repayment behavior varied across lending companies.
- **Higher credit utilization** was consistently associated with increased default risk, making it a key indicator for risk assessment.
- **UPI / Wallet users** demonstrated better repayment performance with fewer missed payments compared to other repayment methods.
- Correlation and hypothesis testing showed that **no single variable strongly explains customer risk**, indicating that BNPL behavior depends on multiple interacting factors.

Machine Learning Analysis

Model Used: Linear Regression

Target Variables

- Loan Amount
- Lending Company
- Age Category
- Credit Utilization Ratio
- Default Risk.

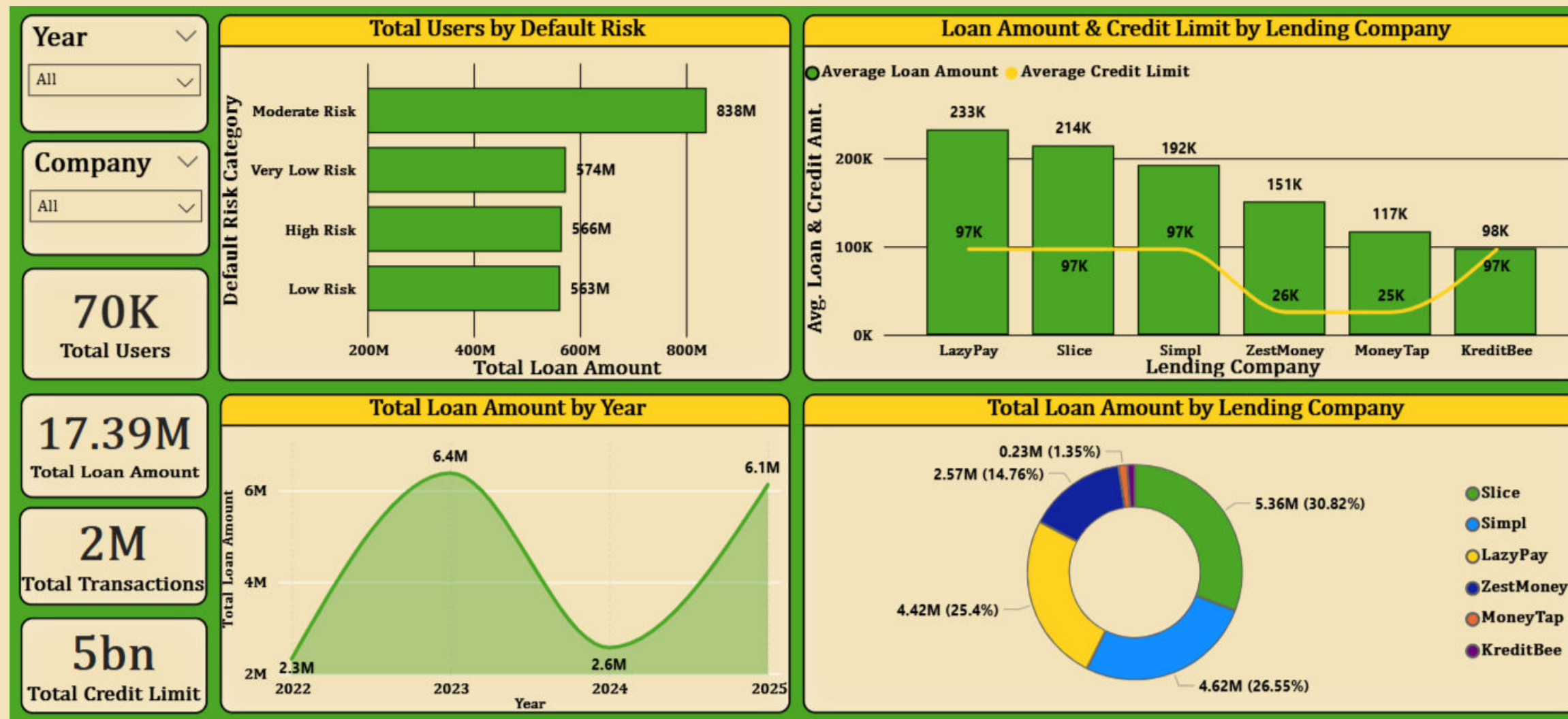
Model Performance (R^2 Scores)

- Loan Amount Model: 0.39 (Explains 39% of the changes).
- Lending Company Model: 0.165 (Explains 16.50% changes).
- Age Category Model: 0.0989 (Explains 9.89% changes).
- Credit Utilization Model: 0.0024 (Explains 0.24% changes).
- Default Risk Model: -0.0002 (No meaningful prediction).

Key Findings

- The Loan Amount model achieved the highest R^2 score (0.39), showing moderate predictive performance compared to the other target variables.
- The Lending Company and Age Category models demonstrated limited predictive capability, while Credit Utilization and Default Risk showed very weak performance.
- The results indicate that Linear Regression can capture some financial relationships but is not suitable for accurately predicting complex customer behavior and risk-related variables.

Executive Overview

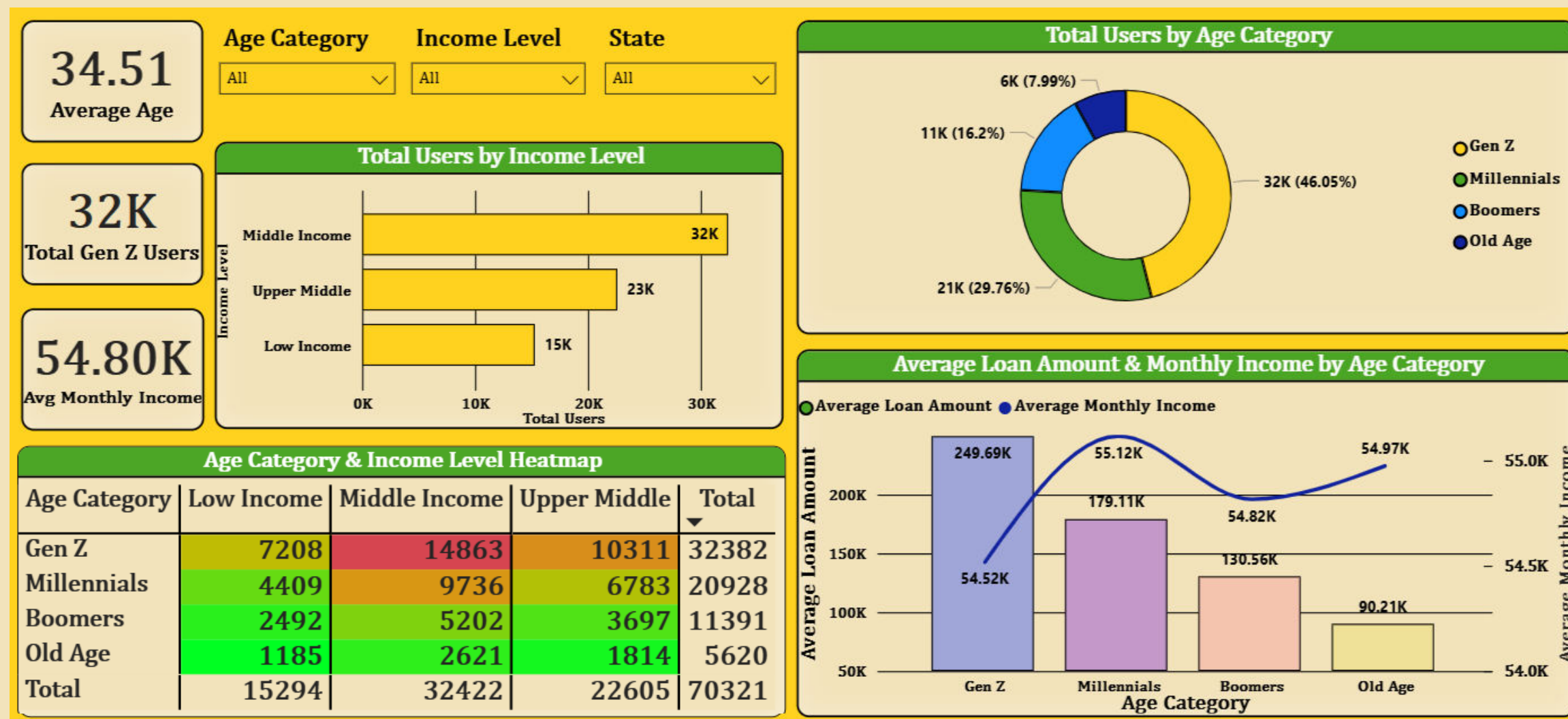


This dashboard presents a high-level summary of the BNPL portfolio using KPIs and trend analysis. It enables quick evaluation of customer growth, lending performance, and overall business health.

Key Findings

- The platform serves 70K customers, with ₹17.39M in total loan amount and 2M transactions.
- Moderate Risk customers contribute the highest loan amount (₹838M) among all risk categories.
- Slice leads the lending portfolio (₹5.36M), followed by Simpl (₹4.62M) and LazyPay (₹4.42M).
- Total loan disbursement peaked in 2023 (₹6.4M), declined in 2024, and recovered to ₹6.1M in 2025.

Customer Demographics Dashboard

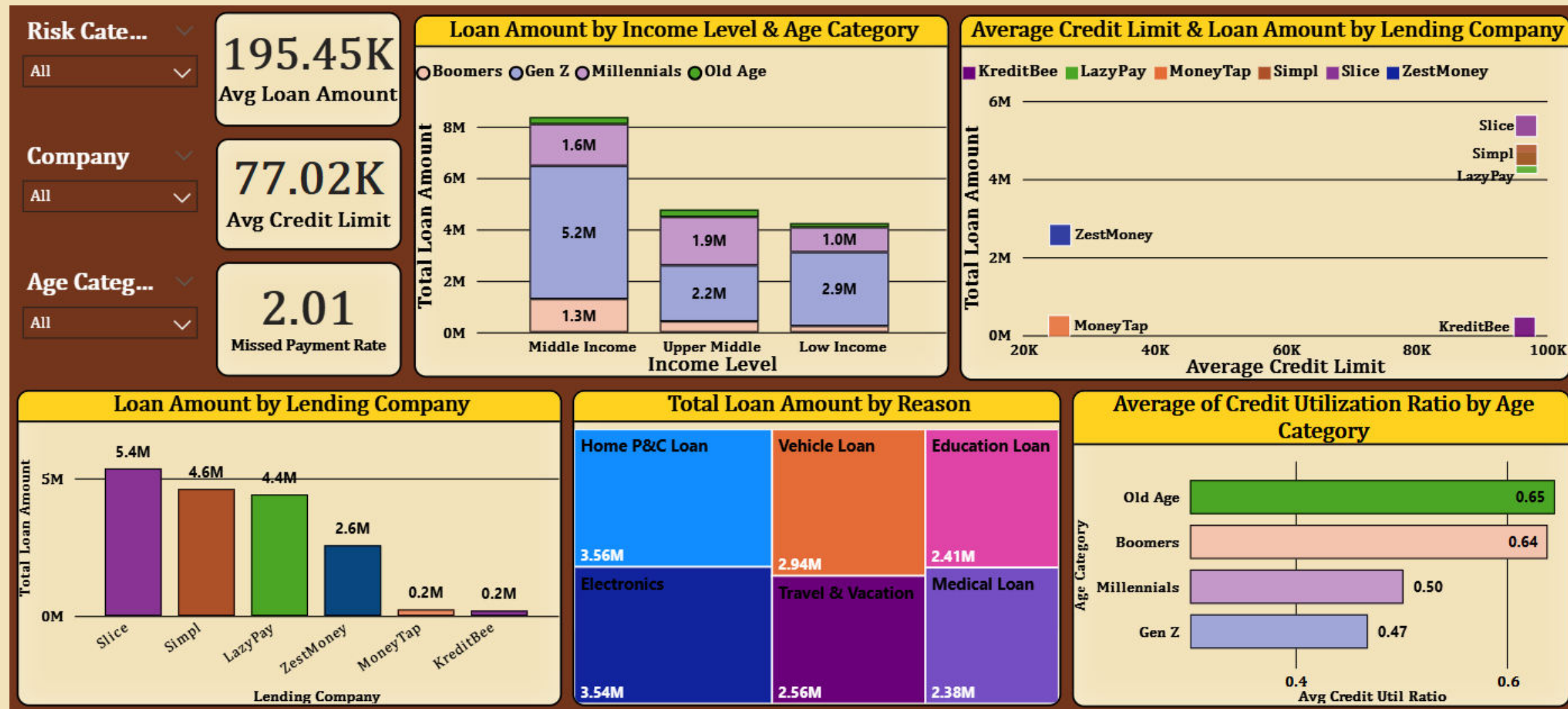


This dashboard analyzes customer distribution across age groups, income levels, and geographic regions to identify the primary users of BNPL services.

Key Findings

- Gen Z is the largest customer segment with 32K users (46.05%).
- Middle-income customers represent the highest user base (32K users).
- Gen Z has the highest average loan amount (₹249.69K).
- Millennials have the highest average monthly income (₹55.12K).

Loan & Credit Analysis

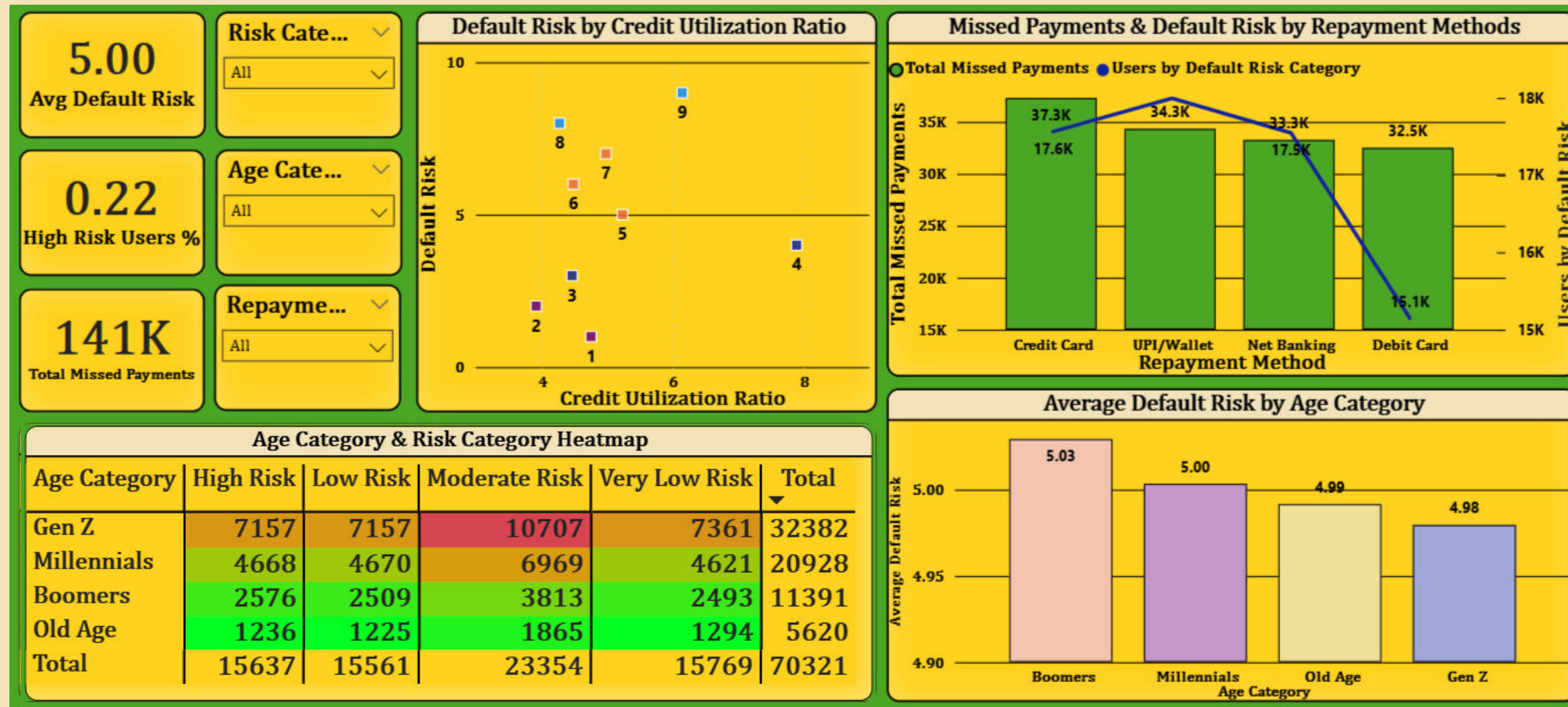


This dashboard examines lending behavior, credit allocation, and borrowing patterns across customer segments and lending companies.

Key Findings

- Slice has the highest total loan amount (₹5.4M), followed by Simpl (₹4.6M) and LazyPay (₹4.4M).
- Home P&C Loans and Electronics Loans are the most common borrowing purposes.
- KreditBee offers the highest average credit limit, while MoneyTap has the lowest.
- Old Age customers have the highest average credit utilization (65%), indicating greater reliance on available credit.

Risk & Repayment Analysis

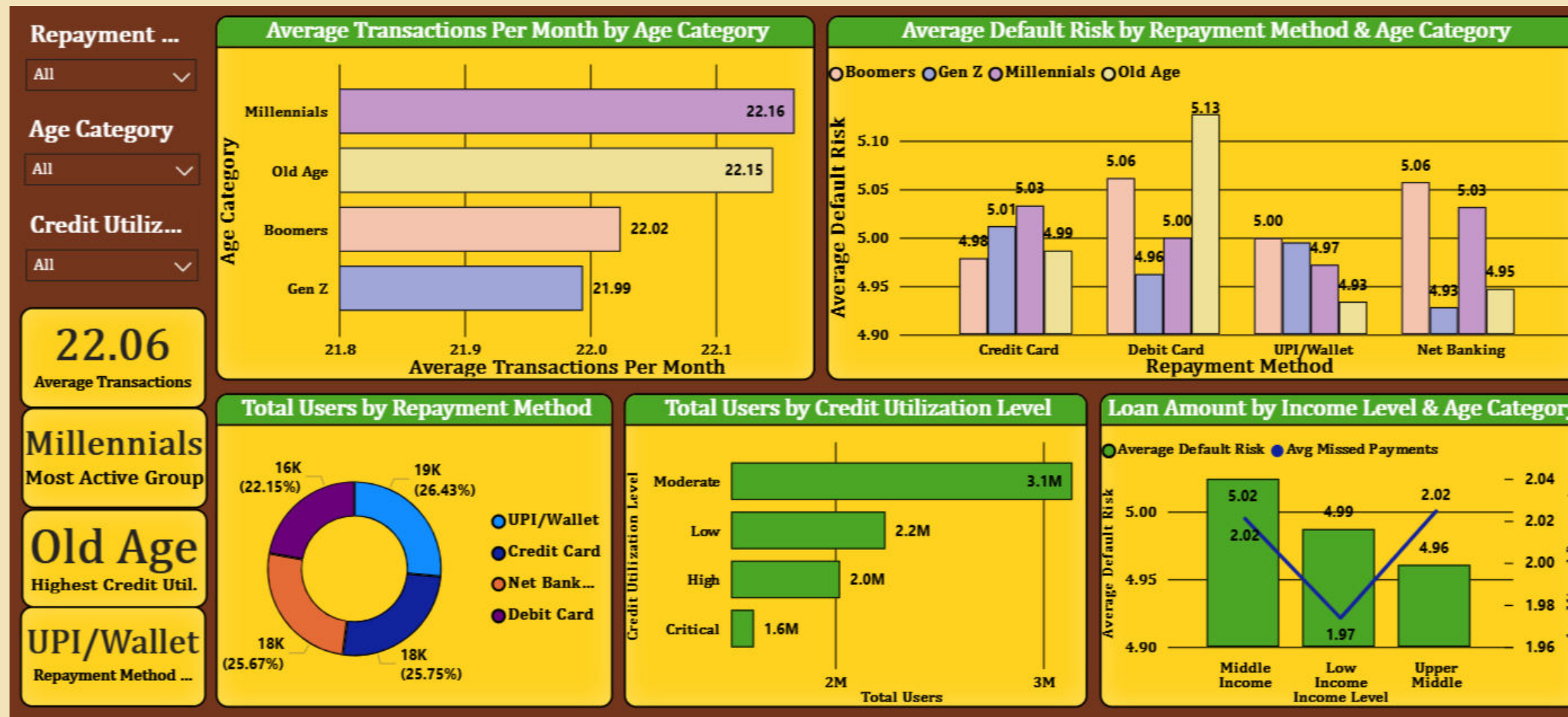


This dashboard focuses on repayment performance, default risk, missed payments, & customer credit behavior to identify high-risk segments.

Key Findings

- The average Default Risk across all customers is 5.00, with 141K total missed payments.
- Higher credit utilization is associated with increased default risk.
- Credit Card users record the highest missed payments (37.3K), while Debit Card users record the lowest (32.5K).
- Average Default risk is lowest in Gen Z but they also show High Risk in Volume.
- Early identification of high-risk customers can significantly improve portfolio & CIBIL score.

Customer Behavior Analysis

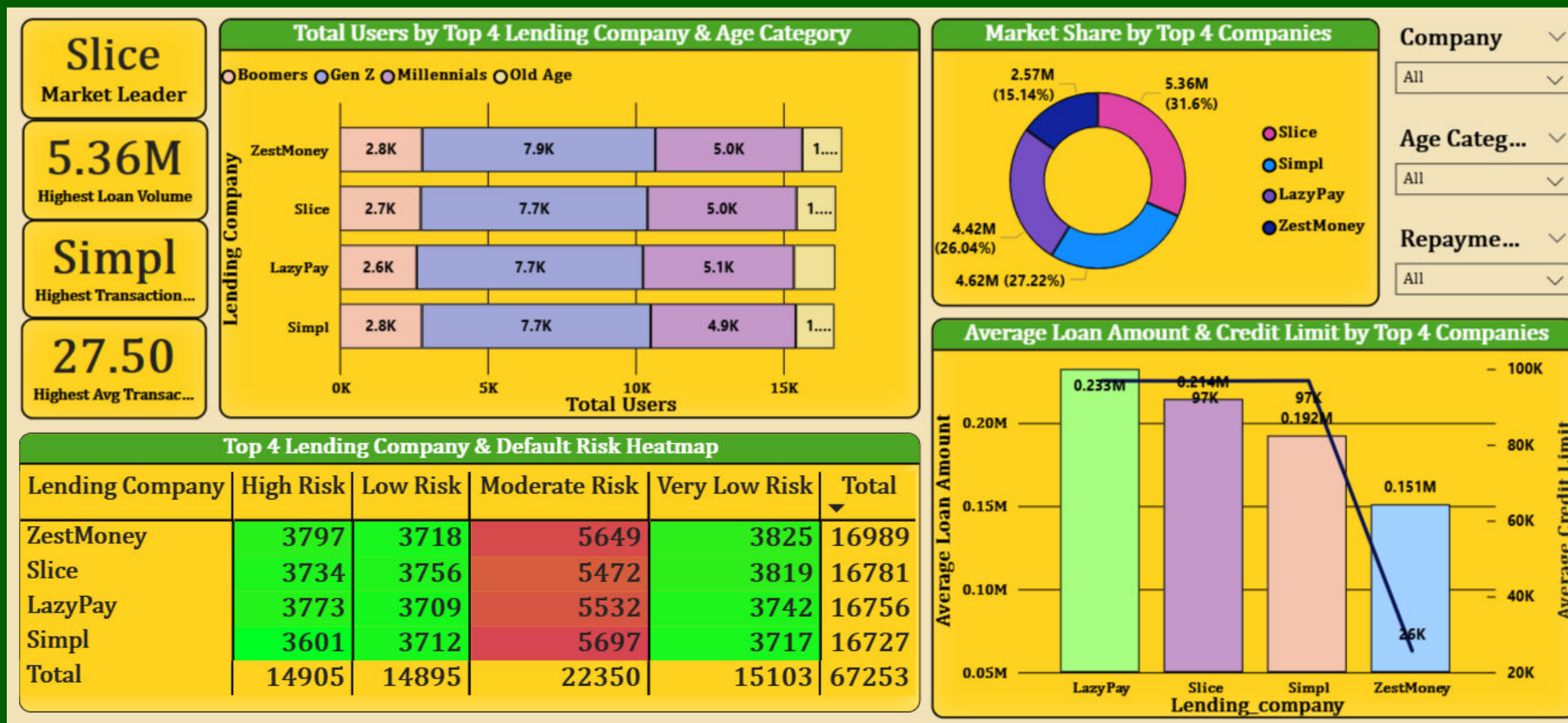


This dashboard analyzes customer spending patterns, repayment preferences, transaction frequency, and credit utilization to better understand user behavior across different customer segments.

Key Findings

- Millennials are the most active users, averaging 22.16 transactions/month.
- UPI/Wallet is the most preferred repayment method (19K users) with comparatively lower default risk.
- Moderate Credit Utilization has the largest customer base (3.1M users).
- Middle-income users show the highest average default risk, while Upper Middle-income users have the highest missed payments.

Competitor Analysis



This dashboard compares the performance of major BNPL providers based on lending volume, customer base, repayment behavior, and portfolio quality. It highlights the competitive positioning of leading players in the Indian BNPL market.

Key Findings

- Slice is the market leader with the highest loan volume (₹5.36M) and 31.6% market share.
- Simpl records the highest average transactions, indicating strong customer engagement.
- LazyPay offers the highest average loan amount (₹233K), while Slice provides the highest average credit limit (₹97K).
- Moderate Risk is the largest customer segment across all leading companies.

Competitor Analysis

Why Leading BNPL Companies Stand Out ?

- **Financial Inclusion:** Expanded access to credit for young users and customers underserved by traditional banks.
- **Alternative Credit Assessment:** Uses behavioral and transaction-based data instead of relying solely on traditional credit scores.
- **Zero-Cost Customer Experience:** Eliminates joining fees, annual charges, and hidden costs to improve customer acquisition.
- **User-Centric Mobile Platform:** Provides a fast, intuitive, and digital-first experience through mobile applications.
- **Flexible Repayment Options:** Offers interest-free EMIs and bill-splitting features to improve affordability.
- **Banking Partnerships:** Collaborates with regulated financial institutions to strengthen lending operations.
- **UPI-Driven Growth:** Leverages UPI payments for seamless transactions and increased customer adoption.

Competitor Analysis

Competitive Advantages & Market Insights

Why Slice Leads the Market ?

- **Gen Z Focus:** Targets students and young professionals, driving high customer adoption.
- **Simple Onboarding:** Quick digital KYC and instant approvals improve accessibility.
- **Flexible BNPL Experience:** Interest-free installments and bill splitting increase convenience.
- **High Customer Engagement:** Rewards and an intuitive app encourage frequent usage.
- **Strong Market Presence:** One of India's leading BNPL platforms with a large active customer base.

Business Advantage

Slice combines a digital-first experience, flexible repayment, and a strong Gen Z customer base, making it one of the fastest-growing BNPL platforms in India.

Competitor Analysis

Emerging Companys' Advantage

How KreditBee Maintains Lower Default Risk ?

- **AI-Based Underwriting:** Evaluates customers using alternative data beyond traditional credit scores.
- **Gradual Credit Limits:** Starts with smaller loans and increases limits based on repayment history.
- **Strict Credit Monitoring:** Reports defaults to credit bureaus, encouraging responsible repayment.
- **Automated Collections:** Uses digital reminders and structured EMI recovery to minimize defaults.

Business Advantage

Slice prioritizes rapid customer growth and engagement as it has first mover advantage & is the most dominant company in this space whereas KreditBee focuses on disciplined credit evaluation & repayment monitoring as it is an Emerging Company & tends to provide an alternative to older companies who have currently captured the market resulting in comparatively lower default risk.

Key Insights & Future of Project

Key Business Insights

- Gen Z contributes the highest share of BNPL users and loan demand.
- High Credit Utilization Ratio is the strongest indicator of potential default risk.
- UPI consistently demonstrates the best repayment performance.
- Slice leads loan volume, while Kreditbee maintains comparatively stronger portfolio quality.
- Credit Utilization Ratio emerged as the strongest indicator of customer risk.
- Customer segmentation enables more accurate lending and marketing strategies.

Future To Dos & Scope

- Deploy advanced ML models such as Random Forest & XGBoost for improved risk prediction.
- Expand into Tier-2 and Tier-3 cities, where BNPL adoption is rapidly increasing.
- Continue monitoring Gen Z, students, & young professionals, fastest-growing segments.
- Integrate real-time transaction monitoring and automated fraud detection.
- Enhance dashboards with forecasting, customer lifetime value, and portfolio health tracking.

Strategic Recommendations

Recommendations for BNPL Providers want to enter this Space

Risk Management:

- Adopt AI-based credit scoring using behavioral and transaction data.
- Monitor customers with high utilization and repeated missed payments.

Customer Growth:

- Focus on Gen Z and middle-income professionals.
- Expand services in Tier-2 and Tier-3 cities with growing digital adoption.

Business Intelligence:

- Implement real-time Business dashboards for continuous monitoring.
- Regularly track business KPIs to improve portfolio performance.

Customer Experience:

- Promote UPI-based repayments through cashback and rewards.
- Provide personalized credit limits based on repayment history.

Conclusion

Project Conclusion

This project demonstrates how data analytics can improve decision-making within the Indian BNPL ecosystem. By combining SQL, Python, statistical techniques, machine learning, and Power BI, raw transactional data can be transformed into meaningful business insights.

Final Takeaways

- Customer demographics, repayment behavior, and credit utilization are key drivers of lending performance.
- Data-driven analytics enables better customer segmentation and risk assessment.
- Interactive dashboards support faster business decisions through real-time KPI monitoring.
- The proposed analytical framework can help financial institutions reduce default risk while improving customer experience and business growth.



Thank You